

GRASSWORKS: Analysis of Bauer et al. (submitted) Functional traits of grasslands: Community weighted mean of canopy height compared to reference sites

MB

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NOTE: To compare different models, you only have to change the models in the section ‘Load models’	

Preparation

Protocol of data exploration (Steps 1-8) used from Zuur et al. (2010) Methods Ecol Evol DOI: 10.1111/2041-210X.12577

```
library(here)
library(tidyverse)
```

Packages

```
## Warning: Paket 'ggplot2' wurde unter R Version 4.5.3 erstellt
## Warning: Paket 'purrr' wurde unter R Version 4.5.3 erstellt
## Warning: Paket 'dplyr' wurde unter R Version 4.5.3 erstellt
```

```
library(ggbeeswarm)
library(patchwork)
library(DHARMA)
library(emmeans)
```

```
sites <- read_csv(
  here("data", "processed", "data_processed_sites_refs.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(
    .default = "?",
    eco.id = "f",
    region = col_factor(levels = c("north", "centre", "south"), ordered = TRUE),
    site.type = col_factor(
      levels = c("positive", "restored", "negative"), ordered = TRUE
    ),
    obs.year = "f",
    hydrology = col_factor(levels = c("moist", "fresh", "dry"), ordered = TRUE),
    mngm.type = "f"
  )
) %>%
  rename(y = cwm.abu.height) %>%
  #filter(y < 1) # see section Outliers: Exclude site N_DAM (more or less only the tall grass Arrhenath
```

Load data

Data exploration

Means and deviations

```
Rmisc::CI(sites$y, ci = .95)
```

```
##      upper      mean      lower
## 0.4842538 0.4730696 0.4618855
```

```
median(sites$y)
```

```
## [1] 0.47
```

```
sd(sites$y)
```

```
## [1] 0.143179
```

```
quantile(sites$y, probs = c(0.05, 0.95), na.rm = TRUE)
```

```
##      5%      95%
```

```
## 0.27 0.70
```

```
sites %>% count(eco.id)
```

```
## # A tibble: 3 x 2
##   eco.id      n
##   <fct>  <int>
## 1 654      209
## 2 664      212
## 3 686      211
```

```
sites %>% count(site.type)
```

```
## # A tibble: 3 x 2
##   site.type      n
##   <ord>    <int>
## 1 positive     57
## 2 restored    472
## 3 negative    103
```

```
sites %>% count(hydrology)
```

```
## # A tibble: 3 x 2
##   hydrology      n
##   <ord>    <int>
## 1 moist     135
## 2 fresh     327
## 3 dry       170
```

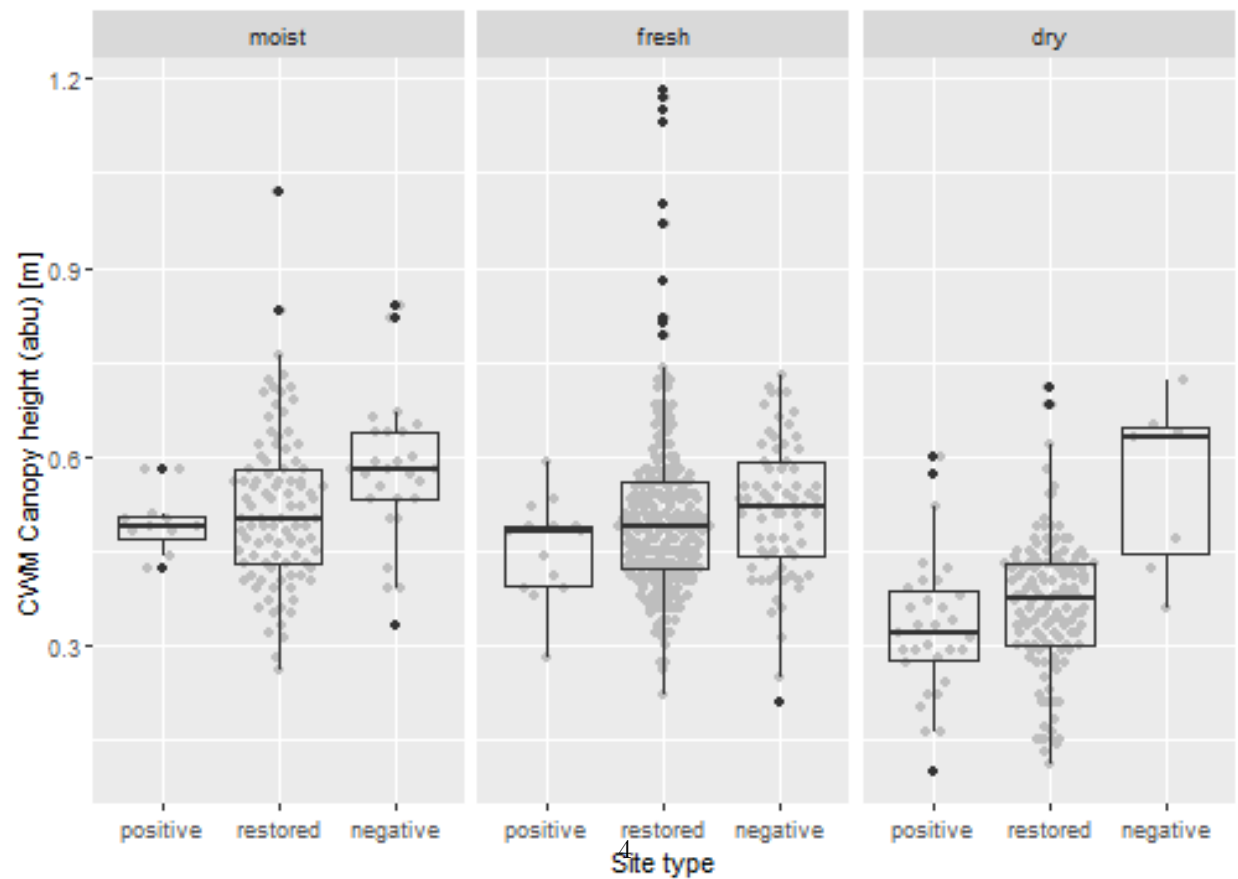
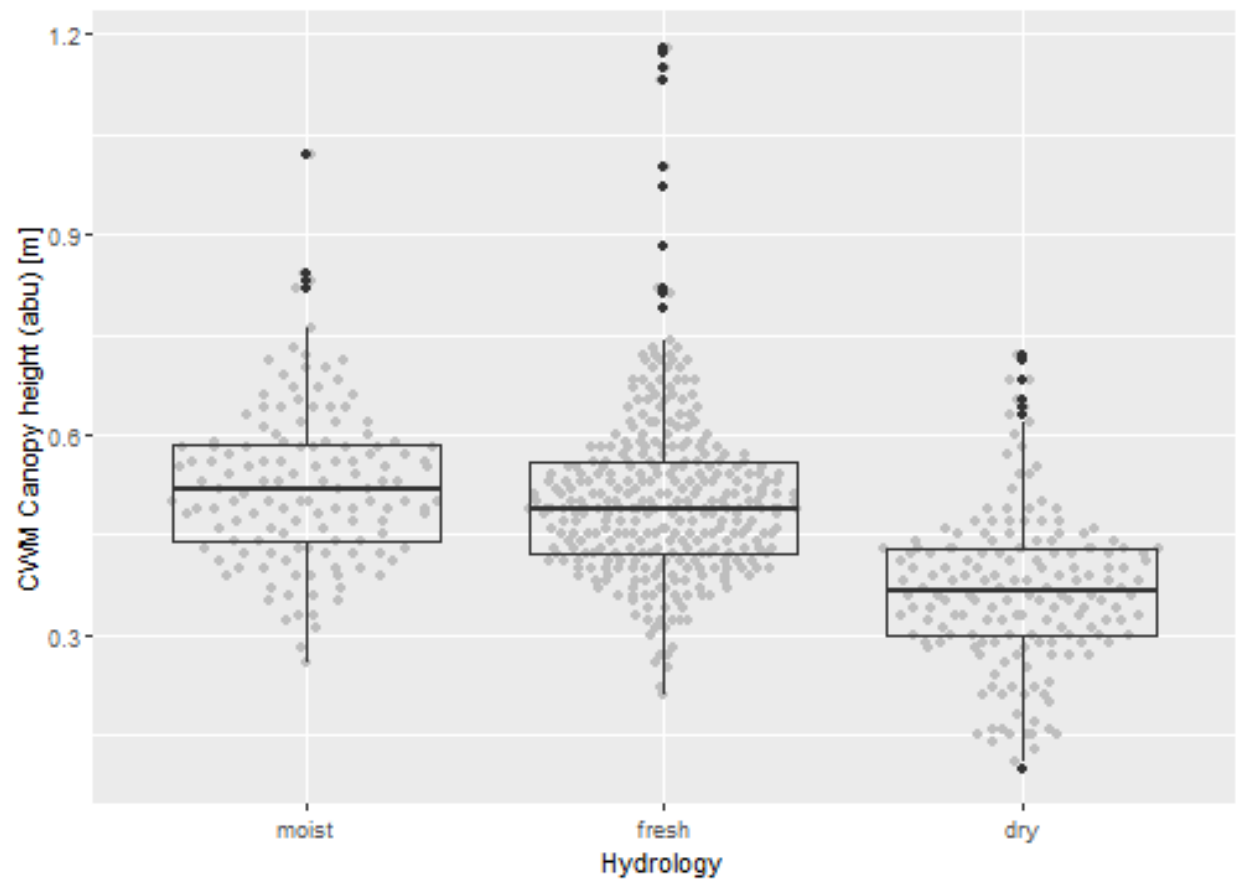
```
sites %>% count(hydrology, eco.id)
```

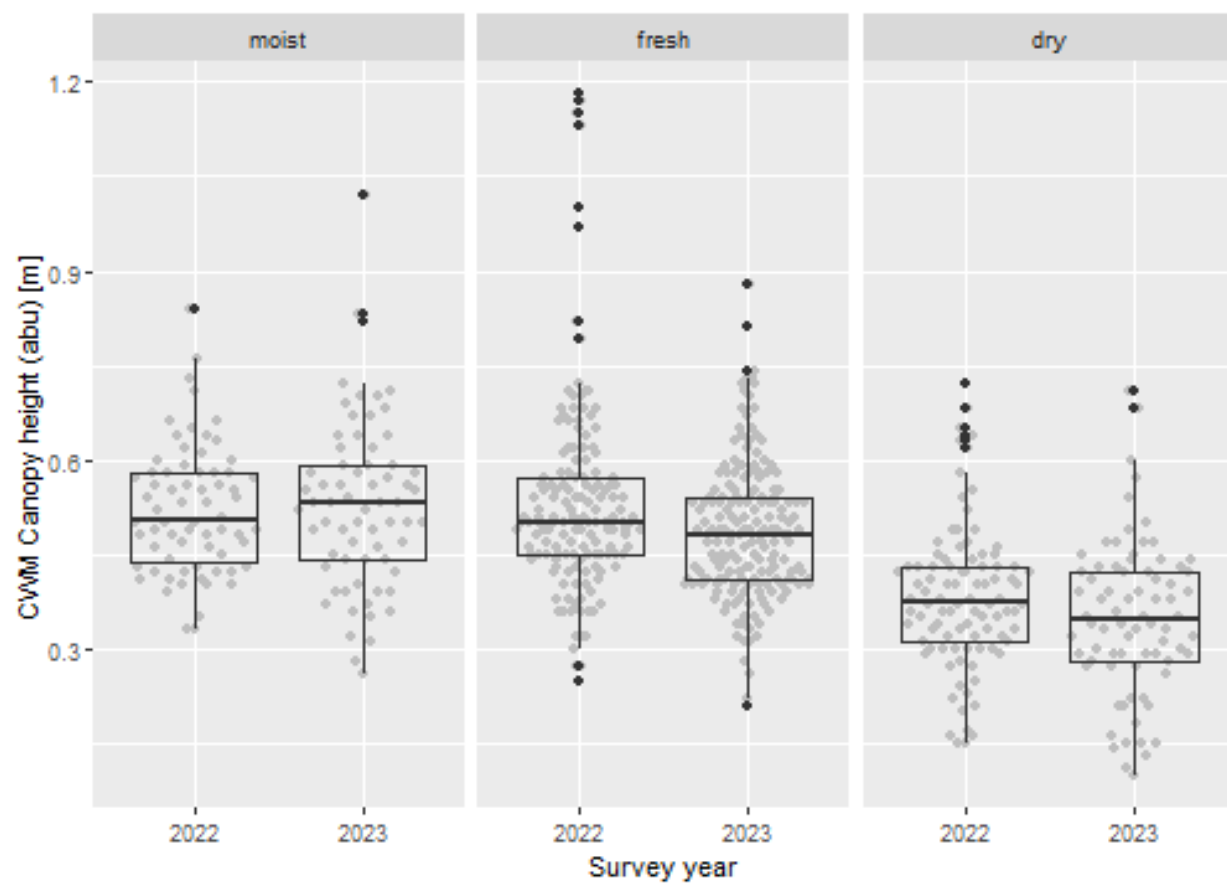
```
## # A tibble: 9 x 3
##   hydrology eco.id      n
##   <ord>    <fct>  <int>
## 1 moist     654      35
## 2 moist     664      50
## 3 moist     686      50
## 4 fresh     654     108
## 5 fresh     664     109
## 6 fresh     686     110
## 7 dry       654      66
## 8 dry       664      53
## 9 dry       686      51
```

```
sites %>% count(hydrology, site.type)
```

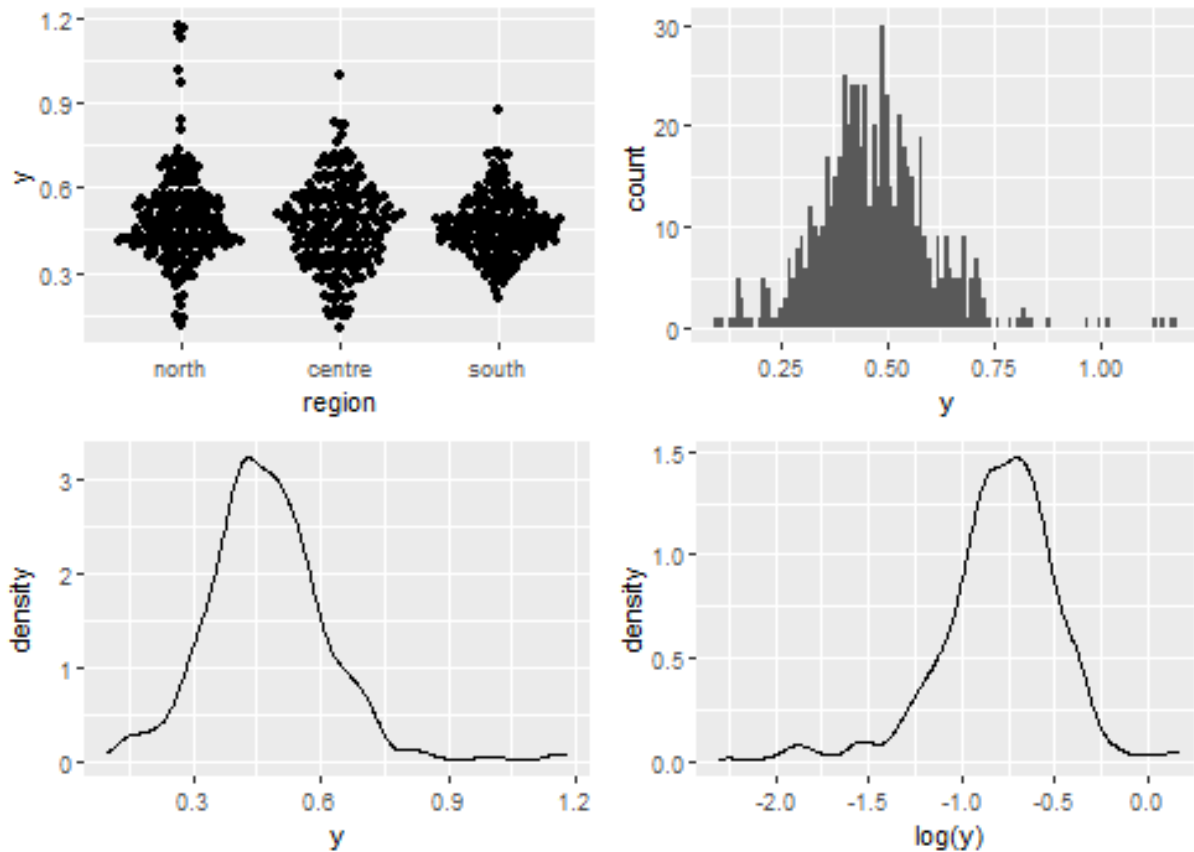
```
## # A tibble: 9 x 3
##   hydrology site.type      n
##   <ord>    <ord>    <int>
## 1 moist     positive     12
## 2 moist     restored     96
## 3 moist     negative     27
## 4 fresh     positive     14
## 5 fresh     restored    244
## 6 fresh     negative     69
## 7 dry       positive     31
## 8 dry       restored    132
## 9 dry       negative      7
```

Graphs of raw data (Step 2, 6, 7)





Outliers, zero-inflation, transformations? (Step 1, 3, 4)



Check collinearity part 1 (Step 5)

Exclude $r > 0.7$ Dormann et al. 2013 *Ecography* DOI: 10.1111/j.1600-0587.2012.07348.x

```
# sites %>%
#   select(where(is.numeric), -y, -starts_with("cum.")) %>%
#   GGally::ggpairs(
#     lower = list(continuous = "smooth_loess")
#   ) +
#   theme(strip.text = element_text(size = 7))

# -> no continuous variables
```

Models

NOTE: Only here you have to modify the script to compare other models

```
load(file = here("outputs", "models", "model_height_refs_1.Rdata"))
load(file = here("outputs", "models", "model_height_refs_2.Rdata"))
m_1 <- m1
m_2 <- m2
```

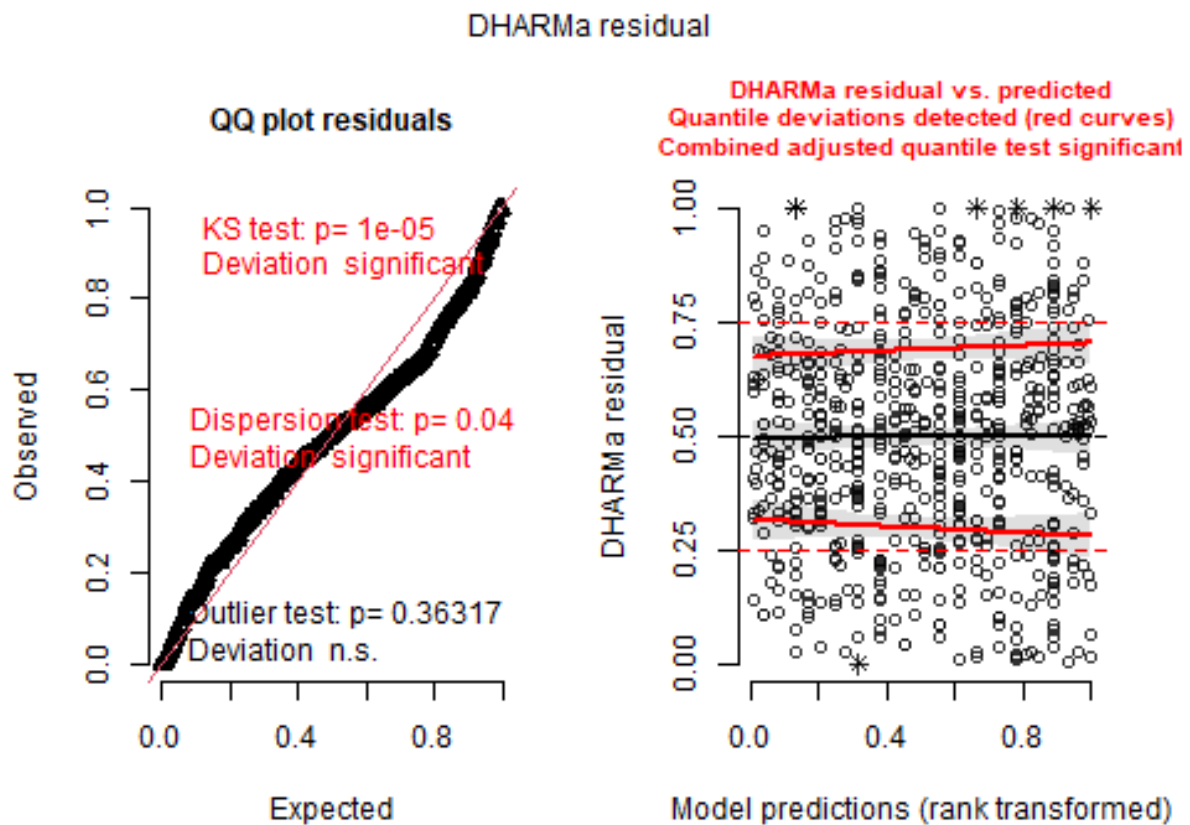
```
m_1@call
## lmer(formula = y ~ hydrology * site.type + eco.id + obs.year +
```

```
##      (1 | id.site), data = sites, REML = FALSE)
m_2@call
## lmer(formula = y ~ (hydrology + site.type + eco.id + obs.year)^2 +
##      (1 | id.site), data = sites, REML = FALSE)
```

Model check

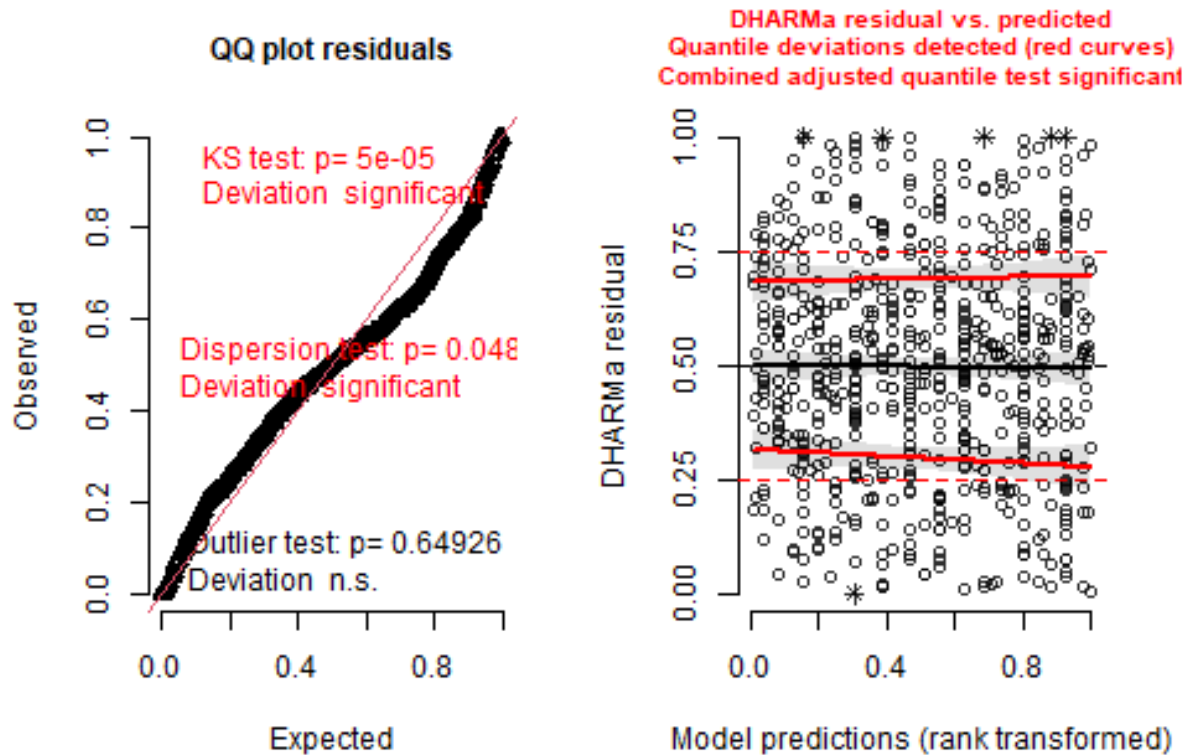
DHARMA

```
simulation_output_1 <- simulateResiduals(m_1, plot = TRUE)
```

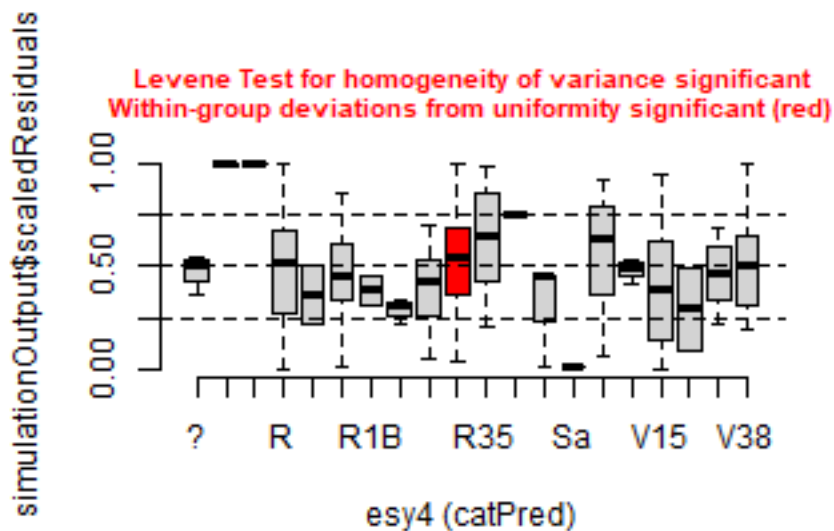


```
simulation_output_2 <- simulateResiduals(m_2, plot = TRUE)
```

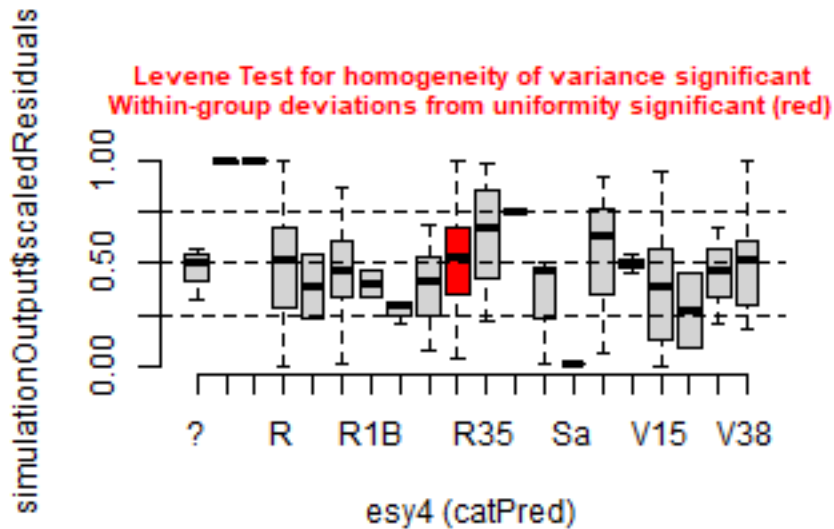
DHARMa residual



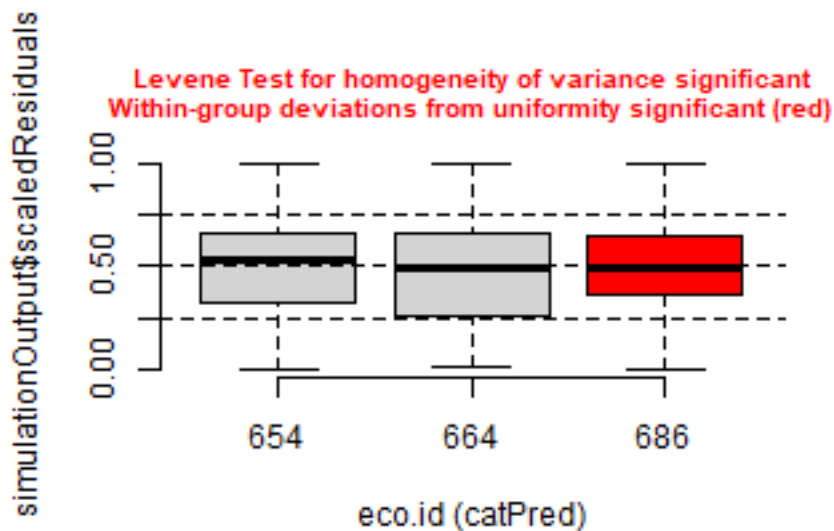
```
plotResiduals(simulation_output_1$scaledResiduals, sites$esy4)
## Warning in ensurePredictor(simulationOutput, predictor):
## DHARMa::ensurePredictor: character string was provided as predictor. DHARMa
## has converted to factor automatically. To remove this warning, please convert
## to factor before attempting to plot with DHARMa.
```



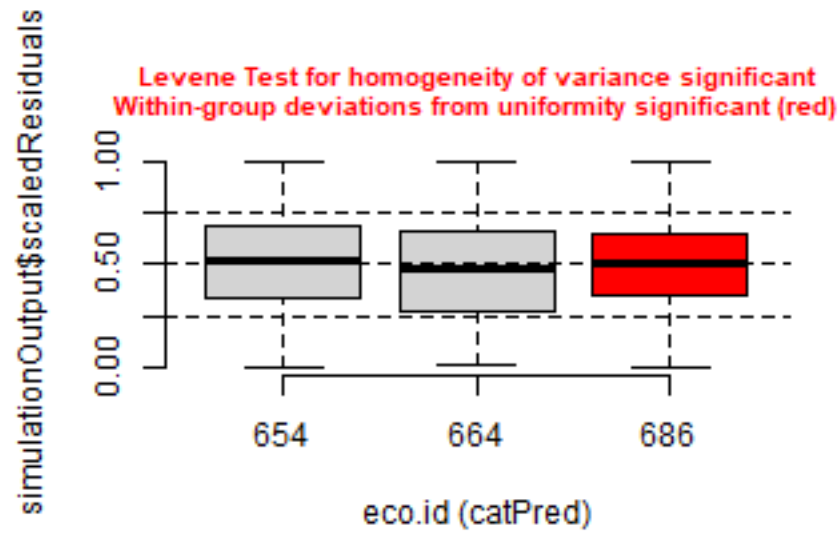

```
plotResiduals(simulation_output_2$scaledResiduals, sites$esy4)
## Warning in ensurePredictor(simulationOutput, predictor):
## DHARMA::ensurePredictor: character string was provided as predictor. DHARMA
## has converted to factor automatically. To remove this warning, please convert
## to factor before attempting to plot with DHARMA.
```



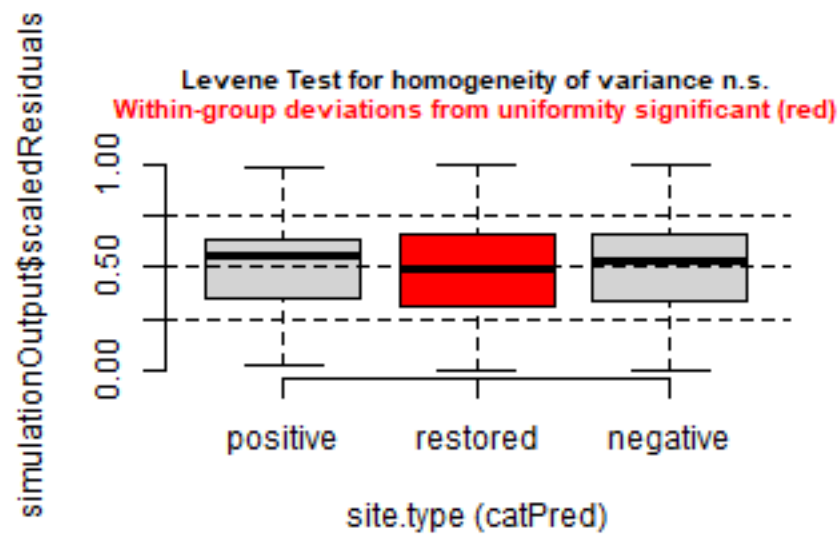
```
plotResiduals(simulation_output_1$scaledResiduals, sites$eco.id)
```



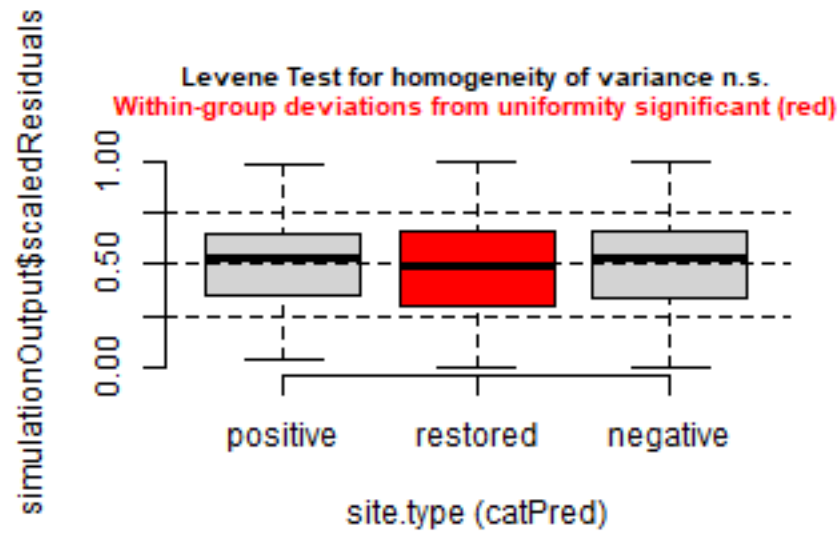
```
plotResiduals(simulation_output_2$scaledResiduals, sites$eco.id)
```



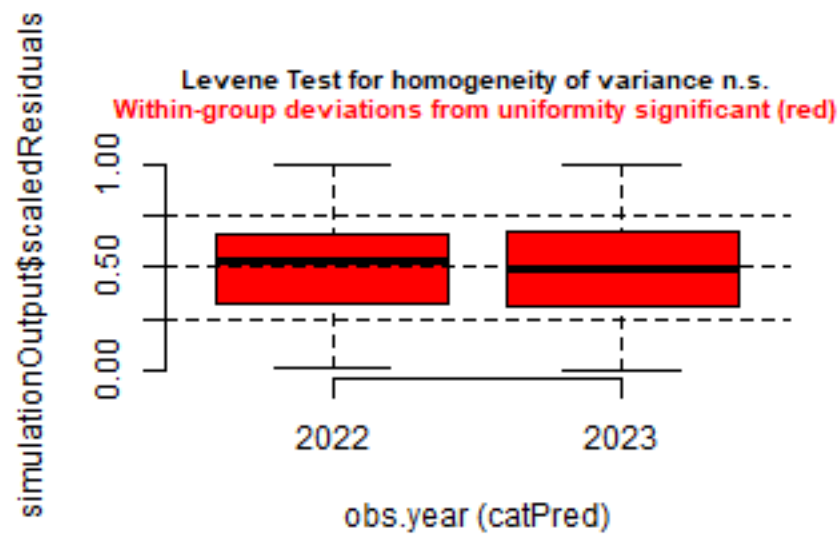
```
plotResiduals(simulation_output_1$scaledResiduals, sites$site.type)
```



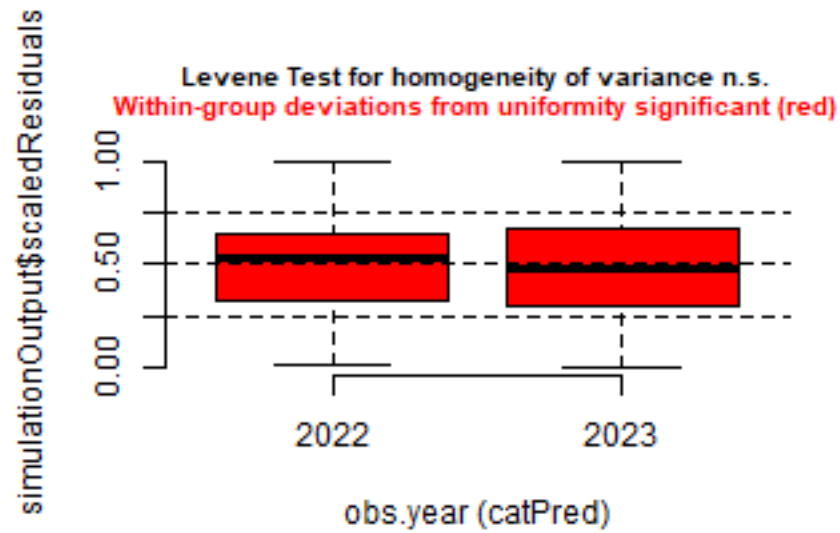
```
plotResiduals(simulation_output_2$scaledResiduals, sites$site.type)
```



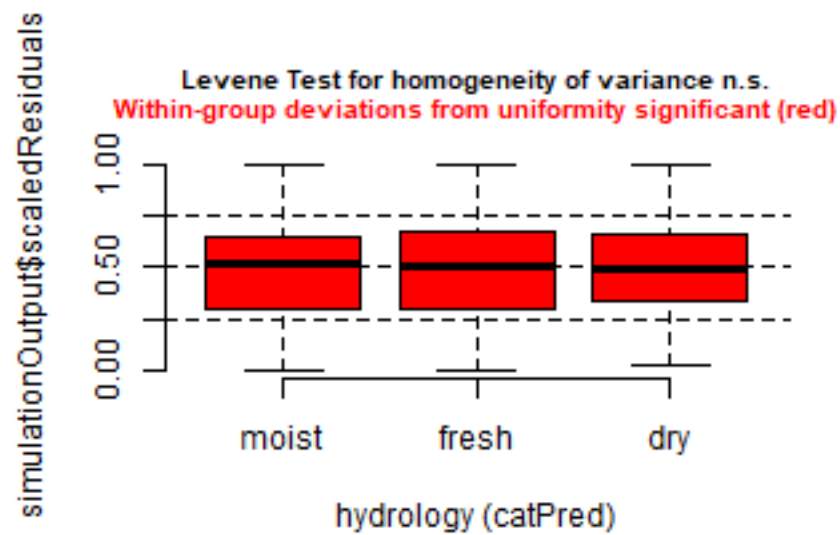
```
plotResiduals(simulation_output_1$scaledResiduals, sites$obs.year)
```



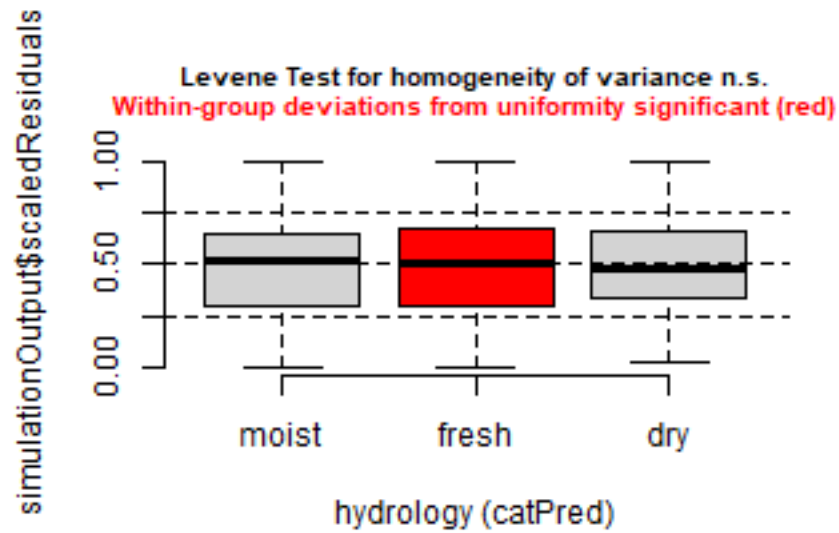
```
plotResiduals(simulation_output_2$scaledResiduals, sites$obs.year)
```



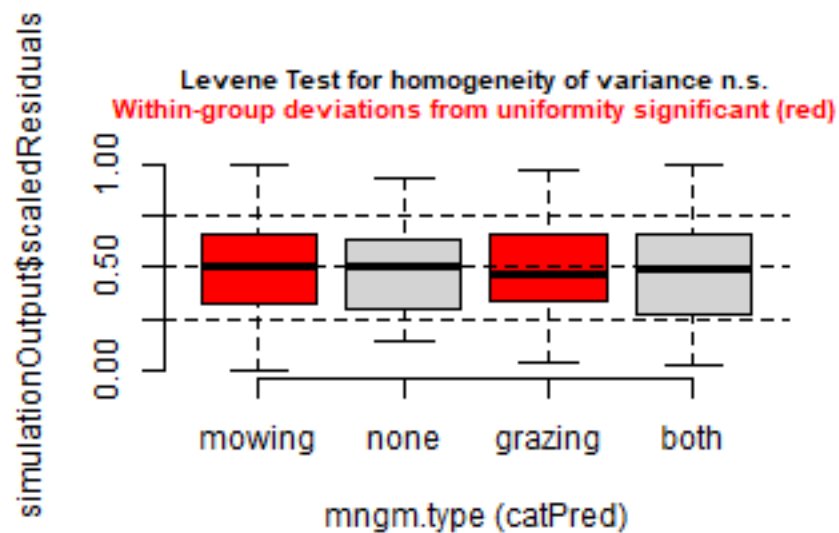
```
plotResiduals(simulation_output_1$scaledResiduals, sites$hydrology)
```



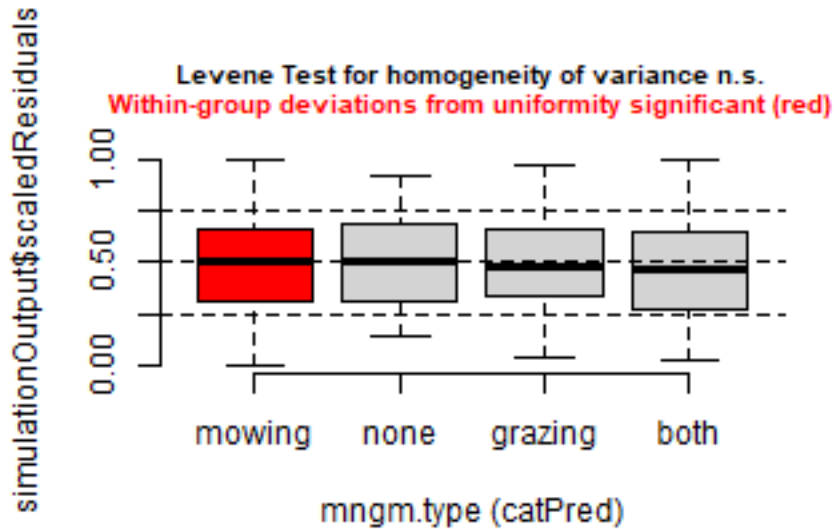
```
plotResiduals(simulation_output_2$scaledResiduals, sites$hydrology)
```



```
plotResiduals(simulation_output_1$scaledResiduals, sites$mngm.type)
```



```
plotResiduals(simulation_output_2$scaledResiduals, sites$mngm.type)
```



Check collinearity part 2 (Step 5)

Remove VIF > 3 or > 10 Zuur et al. 2010 Methods Ecol Evol DOI: 10.1111/j.2041-210X.2009.00001.x

```
car::vif(m_1)
```

```
## Registered S3 method overwritten by 'car':
```

```
## method from
```

```
## na.action.merMod lme4
```

```
##          GVIF Df GVIF^(1/(2*Df))
## hydrology      5.168471  2      1.507789
## site.type      2.214532  2      1.219890
## eco.id         1.038889  2      1.009584
## obs.year       1.059594  1      1.029366
## hydrology:site.type 7.129297  4      1.278294
```

```
car::vif(m_2)
```

```
##          GVIF Df GVIF^(1/(2*Df))
## hydrology     44.347540  2      2.580580
## site.type     30.568689  2      2.351360
## eco.id        9.727224  2      1.766027
## obs.year      4.370346  1      2.090537
## hydrology:site.type 14.352286  4      1.395131
## hydrology:eco.id 15.862927  4      1.412693
## hydrology:obs.year 5.517613  2      1.532632
## site.type:eco.id 29.500872  4      1.526614
## site.type:obs.year 11.380485  2      1.836708
## eco.id:obs.year 9.103046  2      1.736987
```

Model comparison

R² values

```
MuMIn::r.squaredGLMM(m_1)
##           R2m           R2c
## [1,] 0.2532449 0.691834
MuMIn::r.squaredGLMM(m_2)
##           R2m           R2c
## [1,] 0.2904578 0.6917012
```

AICc

Use AICc and not AIC since ratio $n/K < 40$ Burnahm & Anderson 2002 p. 66 ISBN: 978-0-387-95364-9

```
MuMIn::AICc(m_1, m_2) %>%
  arrange(AICc)
##      df      AICc
## m_1 14 -1080.645
## m_2 28 -1063.663
```

Predicted values

Summary table

```
car::Anova(m_1, type = 3)

## Analysis of Deviance Table (Type III Wald chisquare tests)
##
## Response: y
##              Chisq Df Pr(>Chisq)
## (Intercept)    663.1489  1 < 2.2e-16 ***
## hydrology      10.9924  2  0.004102 **
## site.type       9.8379  2  0.007307 **
## eco.id          1.8811  2  0.390408
## obs.year        3.4869  1  0.061857 .
## hydrology:site.type 5.0154  4  0.285716
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

summary(m_1)

## Linear mixed model fit by maximum likelihood ['lmerMod']
## Formula: y ~ hydrology * site.type + eco.id + obs.year + (1 | id.site)
## Data: sites
##
##           AIC          BIC      logLik -2*log(L)  df.resid
##    -1081.3    -1019.0      554.7    -1109.3      618
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.3092 -0.4907 -0.0022  0.4416  4.7386
##
## Random effects:
## Groups Name Variance Std.Dev.
## id.site (Intercept) 0.008811 0.09387
## Residual 0.006191 0.07868
## Number of obs: 632, groups: id.site, 172
```

```
##
## Fixed effects:
##               Estimate Std. Error t value
## (Intercept)      0.49088    0.01906  25.752
## hydrology.L      -0.07856    0.02409  -3.261
## hydrology.Q      -0.02467    0.01977  -1.248
## site.type.L      -0.07919    0.02583  -3.066
## site.type.Q       0.02035    0.01701   1.196
## eco.id664         0.01902    0.01949   0.976
## eco.id686        -0.00667    0.01940  -0.344
## obs.year2023     -0.03022    0.01618  -1.867
## hydrology.L:site.type.L -0.07233    0.04914  -1.472
## hydrology.Q:site.type.L -0.06007    0.04010  -1.498
## hydrology.L:site.type.Q  0.03579    0.03252   1.101
## hydrology.Q:site.type.Q  0.03624    0.02661   1.362
##
## Correlation of Fixed Effects:
##      (Intr) hydr.L hydr.Q st.t.L st.t.Q ec.664 ec.686 o.2023 h.L:...L
## hydrology.L  0.143
## hydrology.Q  0.120  0.202
## site.type.L -0.075 -0.425 -0.397
## site.type.Q  0.486  0.274  0.218 -0.178
## eco.id664   -0.496  0.096 -0.011 -0.072  0.021
## eco.id686   -0.544  0.052  0.038 -0.045 -0.011  0.502
## obs.yer2023 -0.459 -0.033  0.185 -0.034 -0.040 -0.004  0.031
## hydr1.L:...L -0.289 -0.370 -0.327  0.300 -0.383 -0.017  0.059 -0.014
## hydr1.Q:...L -0.297 -0.340  0.089  0.325 -0.366 -0.067  0.029  0.100  0.241
## hydr1.L:...Q  0.205  0.710  0.196 -0.380  0.200  0.042 -0.020 -0.070 -0.338
## hydr1.Q:...Q  0.060  0.205  0.728 -0.362  0.320 -0.018  0.048  0.173 -0.296
##      h.Q:...L h.L:...Q
## hydrology.L
## hydrology.Q
## site.type.L
## site.type.Q
## eco.id664
## eco.id686
## obs.yer2023
## hydr1.L:...L
## hydr1.Q:...L
## hydr1.L:...Q -0.313
## hydr1.Q:...Q  0.083  0.138
```

Forest plot

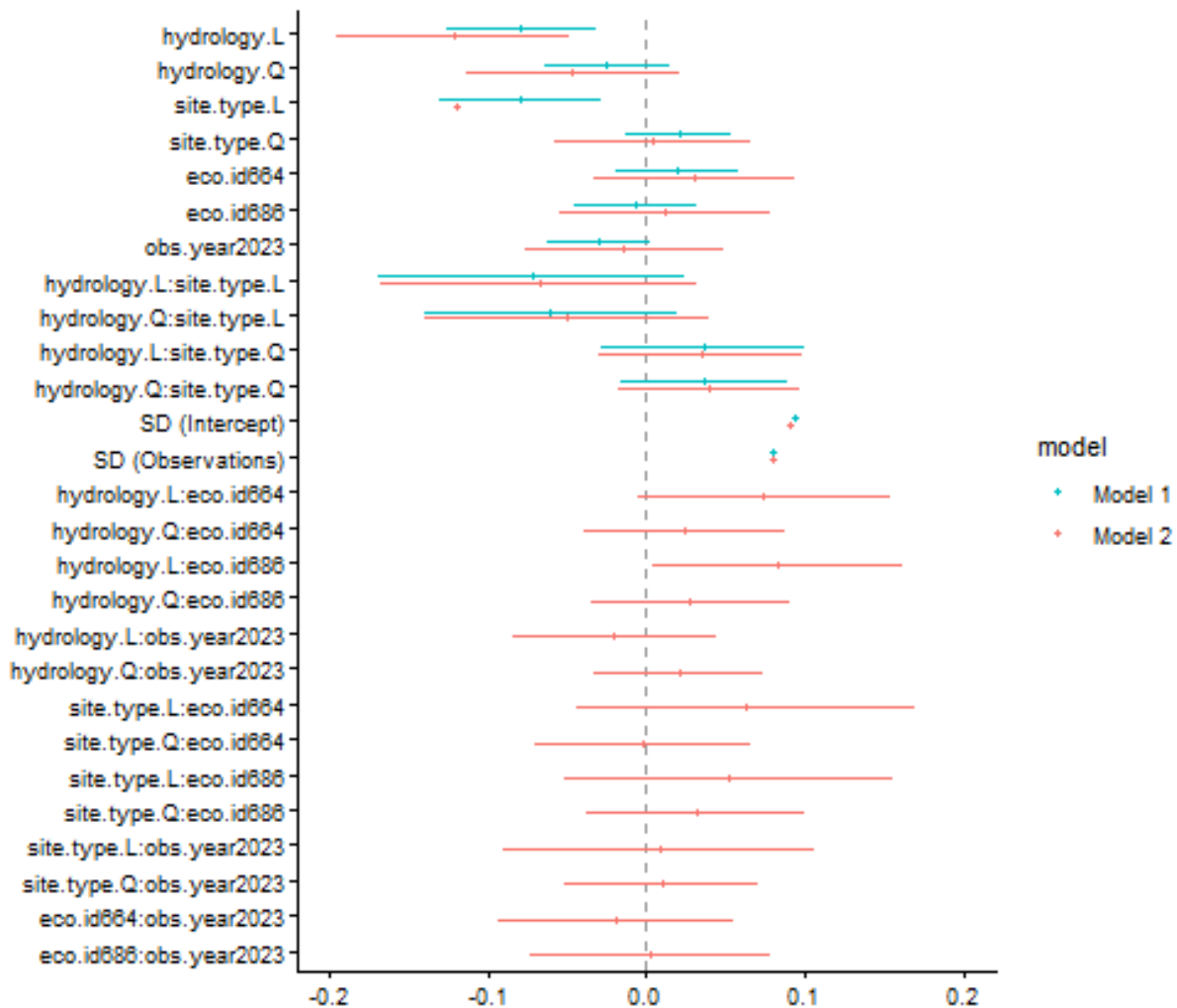
```
dotwhisker::dwplot(
  list(m_1, m_2),
  ci = 0.95,
  show_intercept = FALSE,
  vline = geom_vline(xintercept = 0, colour = "grey60", linetype = 2)) +
  xlim(-0.2, 0.2) +
  theme_classic()
```

```
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.
```



```
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.

## Warning: Using the `size` aesthetic with geom_segment was deprecated in ggplot2 3.4.0.
## i Please use the `linewidth` aesthetic instead.
## i The deprecated feature was likely used in the dotwhisker package.
##   Please report the issue at <https://github.com/fsolt/dotwhisker/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

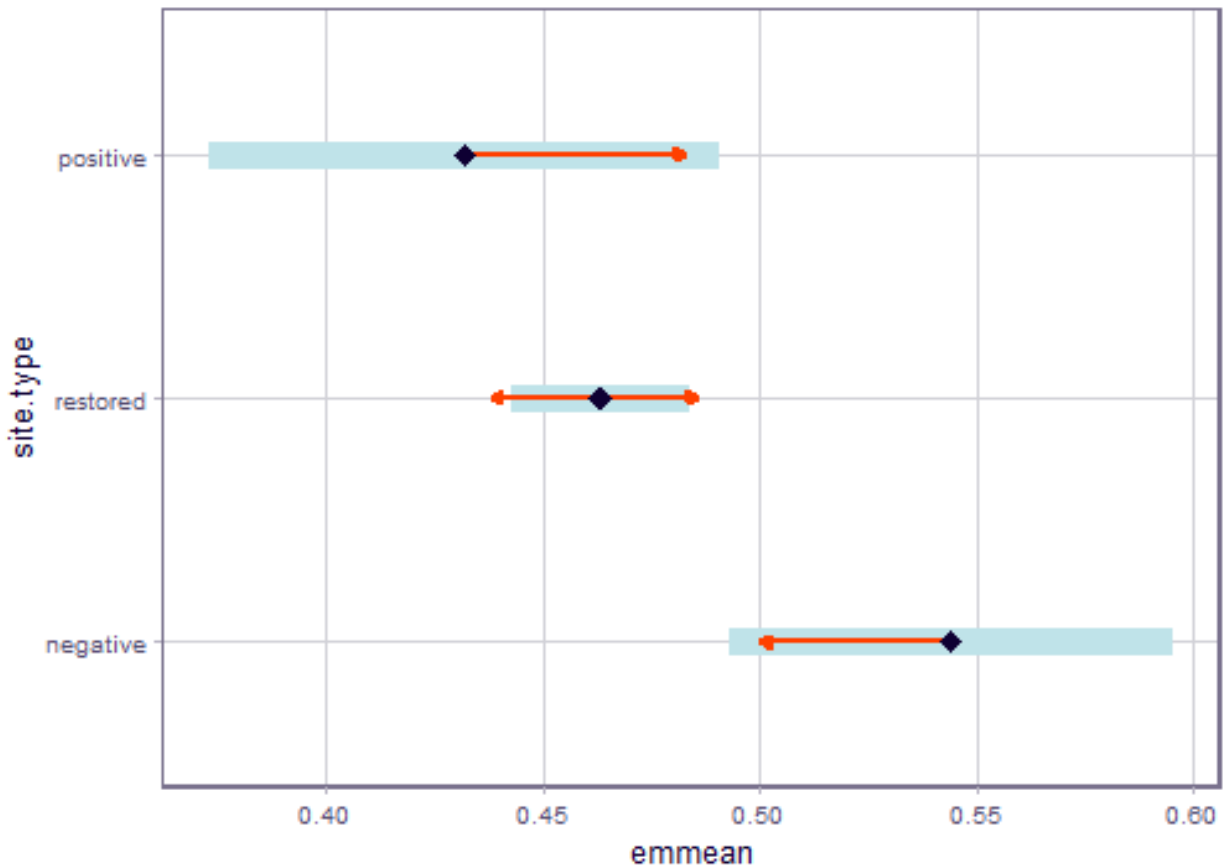


Effect sizes: Hydrology

```
# emm <- emmeans(
#   m_1,
#   reupairwise ~ hydrology,
#   type = "response"
# )
# plot(emm, comparison = TRUE)
# emm
```

Effect sizes: Site type

```
emm <- emmeans(
  m_1,
  revpairwise ~ site.type,
  type = "response"
)
plot(emm, comparison = TRUE)
```



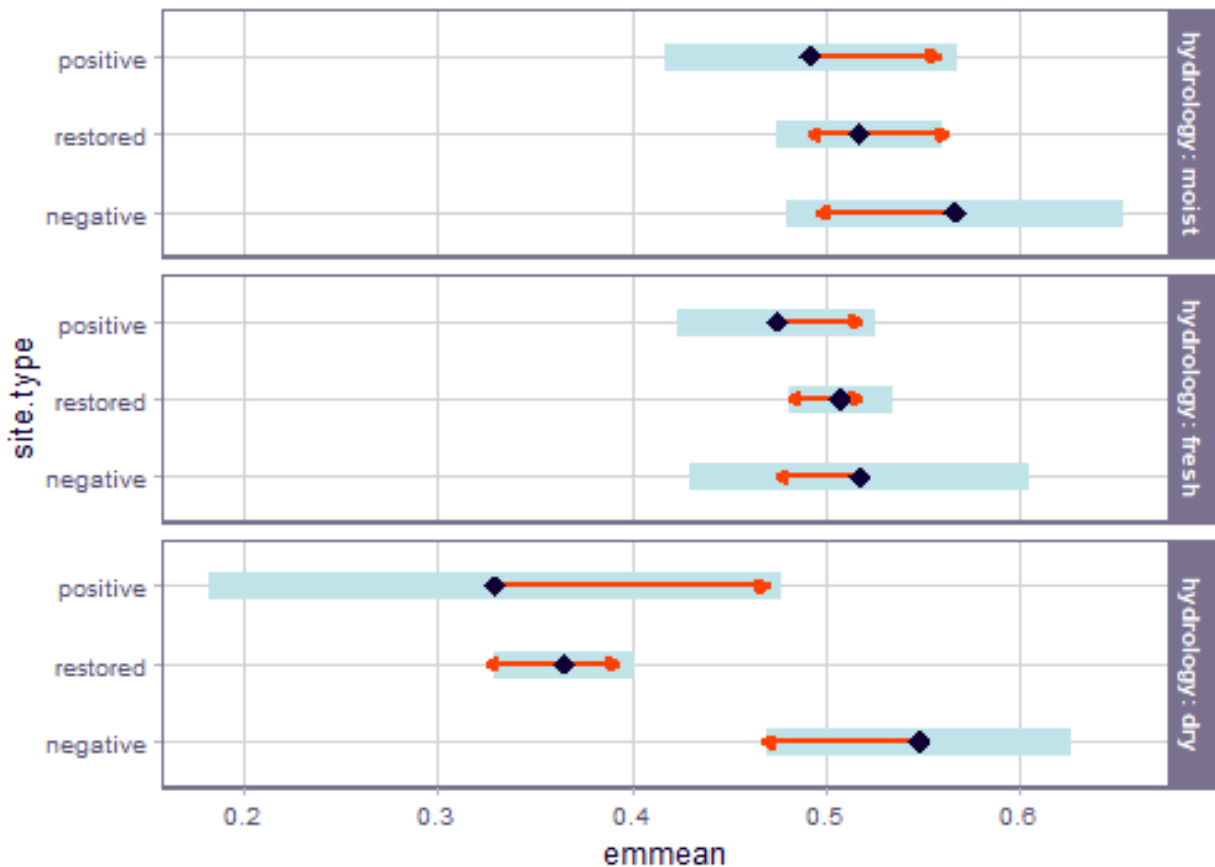
```
emm
```

```
## $emmeans
##   site.type emmean      SE    df lower.CL upper.CL
##   negative  0.544 0.0238  14.1   0.493   0.595
##   restored  0.463 0.0105 177.1   0.443   0.484
##   positive  0.432 0.0293  51.9   0.373   0.491
##
## Results are averaged over the levels of: hydrology, eco.id, obs.year
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
##   contrast      estimate      SE    df t.ratio p.value
##   restored - negative -0.0809 0.0259 16.7  -3.119  0.0165
##   positive - negative -0.1120 0.0377 21.6  -2.967  0.0190
##   positive - restored -0.0311 0.0311 60.7  -0.999  0.5806
```

```
##
## Results are averaged over the levels of: hydrology, eco.id, obs.year
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 3 estimates
```

Effect sizes: Hydrology x site type

```
emm <- emmeans(
  m_1,
  revpairwise ~ site.type | hydrology,
  type = "response"
)
plot(emm, comparison = TRUE)
```



```
emm

## $emmeans
## hydrology = moist:
##   site.type emmean      SE      df lower.CL upper.CL
##   negative  0.567 0.0445 463.11   0.479   0.654
##   restored  0.517 0.0216 176.95   0.475   0.560
##   positive  0.492 0.0381  97.25   0.417   0.568
##
## hydrology = fresh:
##   site.type emmean      SE      df lower.CL upper.CL
##   negative  0.518 0.0447 2802.00   0.430   0.605
```

```
## restored    0.508 0.0135 177.07    0.481    0.534
## positive    0.475 0.0241  15.33    0.424    0.526
##
## hydrology = dry:
## site.type emmean      SE      df lower.CL upper.CL
## negative   0.548 0.0345    8.31    0.470    0.627
## restored   0.365 0.0184 177.01    0.329    0.401
## positive   0.329 0.0756 3928.66    0.181    0.478
##
## Results are averaged over the levels of: eco.id, obs.year
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## hydrology = moist:
## contrast      estimate      SE      df t.ratio p.value
## restored - negative -0.04922 0.0494  346.46  -0.996  0.5796
## positive - negative -0.07434 0.0584  160.73  -1.272  0.4129
## positive - restored -0.02512 0.0438  108.37  -0.573  0.8348
##
## hydrology = fresh:
## contrast      estimate      SE      df t.ratio p.value
## restored - negative -0.00999 0.0467  958.04  -0.214  0.9751
## positive - negative -0.04263 0.0503   24.38  -0.847  0.6780
## positive - restored -0.03263 0.0276   17.29  -1.182  0.4792
##
## hydrology = dry:
## contrast      estimate      SE      df t.ratio p.value
## restored - negative -0.18355 0.0390    9.08  -4.702  0.0028
## positive - negative -0.21901 0.0832   11.82  -2.632  0.0537
## positive - restored -0.03546 0.0778 1189.00  -0.456  0.8918
##
## Results are averaged over the levels of: eco.id, obs.year
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 3 estimates
```

Session info

```
## R version 4.5.2 (2025-10-31 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: Europe/Berlin
## tzcode source: internal
##
```

```

## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] emmeans_2.0.3   DHARMA_0.5.0   patchwork_1.3.2 ggbeeswarm_0.7.3
## [5] lubridate_1.9.5 forcats_1.0.1   stringr_1.6.0   dplyr_1.2.1
## [9] purrr_1.2.2     readr_2.2.0    tidyr_1.3.2     tibble_3.3.1
## [13] ggplot2_4.0.3   tidyverse_2.0.0 here_1.0.2
##
## loaded via a namespace (and not attached):
## [1] Rdpack_2.6.6      gridExtra_2.3      sandwich_3.1-1
## [4] rlang_1.2.0       magrittr_2.0.5     multcomp_1.4-30
## [7] otel_0.2.0        compiler_4.5.2     mgcv_1.9-3
## [10] vctr_0.7.3        pkgconfig_2.0.3    crayon_1.5.3
## [13] fastmap_1.2.0     backports_1.5.1    labeling_0.4.3
## [16] utf8_1.2.6        ggstance_0.3.7     promises_1.5.0
## [19] rmarkdown_2.31    tzdb_0.5.0         nloptr_2.2.1
## [22] bit_4.6.0         xfun_0.58          later_1.4.8
## [25] broom_1.0.13      parallel_4.5.2     R6_2.6.1
## [28] gap.datasets_0.0.6 stringi_1.8.7       qgam_2.0.0
## [31] RColorBrewer_1.1-3 car_3.1-5           boot_1.3-32
## [34] estimability_1.5.1 Rcpp_1.1.1-1.1     iterators_1.0.14
## [37] knitr_1.51        zoo_1.8-15         parameters_0.29.1
## [40] httpuv_1.6.17     Matrix_1.7-4       splines_4.5.2
## [43] timechange_0.4.0  tidysselect_1.2.1  rstudioapi_0.18.0
## [46] abind_1.4-8       yaml_2.3.12        MuMIn_1.48.19
## [49] doParallel_1.0.17 codetools_0.2-20    lattice_0.22-7
## [52] plyr_1.8.9        bayestestR_0.18.1  shiny_1.13.0
## [55] withr_3.0.2       S7_0.2.2           evaluate_1.0.5
## [58] marginaleffects_0.32.0 survival_3.8-3     pillar_1.11.1
## [61] gap_1.15.2        carData_3.0-6      foreach_1.5.2
## [64] stats4_4.5.2      insight_1.5.1      reformulas_0.4.4
## [67] generics_0.1.4    vroom_1.7.1        rprojroot_2.1.1
## [70] hms_1.1.4         scales_1.4.0       minqa_1.2.8
## [73] xtable_1.8-8      glue_1.8.1         tools_4.5.2
## [76] data.table_1.18.4 lme4_2.0-1         mvtnorm_1.4-0
## [79] grid_4.5.2        rbibutils_2.4.1    datawizard_1.3.1
## [82] nlme_3.1-168      Rmisc_1.5.1        performance_0.17.0
## [85] beeswarm_0.4.0    vipor_0.4.7        Formula_1.2-5
## [88] cli_3.6.6         gtable_0.3.6       digest_0.6.39
## [91] pbkrtest_0.5.5    TH.data_1.1-5      farver_2.1.2
## [94] htmltools_0.5.9   lifecycle_1.0.5    mime_0.13
## [97] bit64_4.8.2       dotwhisker_0.8.4   MASS_7.3-65

```